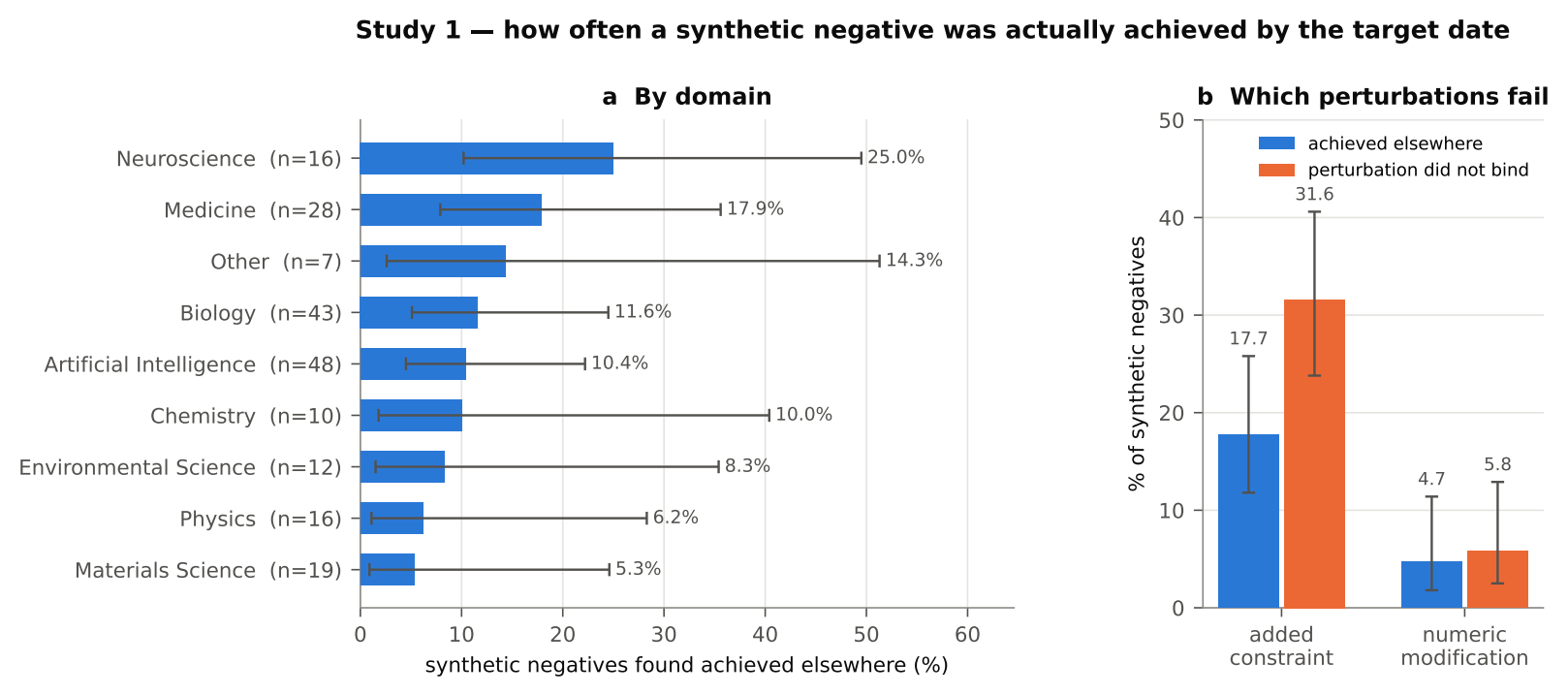
Study 1 — how often a synthetic negative was actually achieved by the target date
a By domain
Neuroscience (n=16)
Medicine (n=28)
Other (n=7)
Biology (n=43)
Artificial Intelligence (n=48)
Chemistry (n=10)
Environmental Science (n=12)
Physics (n=16)
Materials Science (n=19)
25.0%
17.9%
14.3%
11.6%
10.4%
10.0%
8.3%
6.2%
5.3%
0
10
20
30
40
50
60
synthetic negatives found achieved elsewhere (%)
b Which perturbations fail
achieved elsewhere
perturbation did not bind
% of synthetic negatives
0
10
20
30
40
50
17.7
31.6
4.7
5.8
added constraint
numeric modification